# Escaping Saddle Points with Adaptive Gradient Methods

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### The paper in one minute

**Question:** Can adaptive gradient preconditioning make the natural stochastic noise more effective for saddle escape?
**Model/Method:** Idealized adaptive methods update $x_{t+1} = x_t - \eta A_t g_t$, where $A_t$ is estimated from stochastic-gradient second moments, as in a full-matrix RMSProp-like method.
**Finding:** Near stationarity, an inverse-square-root preconditioner can approximately whiten gradient noise; the authors give a second-order convergence analysis for an estimable RMSProp-style variant.
**Why we care:** Adaptive methods can reshape the *directions of noise*, not just rescale parameter learning rates. This is a mechanism distinct from explicitly injected perturbations.

## 1. Noise geometry can help or hinder escape

Stochastic gradient $g_t$ satisfies $\mathbb{E}[g_t] = \nabla f(x_t)$. A preconditioned update has the form

$$x_{t+1} = x_t - \eta A_t g_t = x_t - \eta A_t \nabla f(x_t) - \eta A_t (g_t - \nabla f(x_t)).$$

The first term is the deterministic descent component; the second is noise. Near a saddle with small mean gradient, the noise term can dominate the escape behavior. If stochastic variability is negligible along the most-negative curvature direction, ordinary SGD may escape slowly.

### What does a preconditioner do?

A positive-definite matrix $A$ can scale different directions differently. Full-matrix RMSProp-style preconditioning uses

$$A = (G + \epsilon I)^{-1/2}, \qquad G = \mathbb{E}[gg^\top].$$

This $G$ is the second moment of the stochastic gradients, not always their covariance: $G = \mathrm{Cov}(g) + \nabla f \nabla f^\top$.

### Worked example: whitening two uneven noise directions

At a stationary point suppose a sampled gradient has mean zero and covariance

$$G = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}.$$

The raw noise standard deviations are 3 and 1. With negligible damping,

$$A = G^{-1/2} = \begin{pmatrix} 1/3 & 0 \\ 0 & 1 \end{pmatrix}, \quad AGA^\top = I.$$

After preconditioning, both directions have unit noise variance. Thus a previously under-explored negative-curvature direction can receive substantial stochastic excitation. This is a *toy illustration*, not proof that any particular Adam implementation achieves isotropic noise.

## 2. From idealized preconditioning to an actual algorithm

### Part A: the idealized adaptive method (IAM)

The paper first analyzes what happens *if* an appropriate preconditioner is available. Its convergence constants depend on how $A$ affects the descent direction, gradient-noise magnitude, and the smallest eigenvalue of preconditioned stochastic second moments. Definition 4.2 summarizes these properties with parameters $\Lambda_1, \Lambda_2, \Gamma, \nu, \lambda_-$. These replace a universal claim that every adaptive rule helps.

### Part B: estimating the matrix

The algorithm must estimate $G = \mathbb{E}[gg^\top]$ online. In an RMSProp-style exponential estimate,

$$\widehat{G}_t = \beta \widehat{G}_{t-1} + (1-\beta) g_t g_t^\top, \quad \widehat{A}_t = (\widehat{G}_t + \epsilon I)^{-1/2}.$$

The analysis includes a *burn-in* to estimate a usable preconditioner, and extra care when the algorithm makes larger steps. A large parameter move changes the gradient distribution and can make a stale estimate unreliable.

### The main guarantee—and its scope

Theorems 4.3–4.4 analyze the idealized/noisy-preconditioner procedure, and Corollary 4.2 converts the analysis to a carefully specified RMSProp version. In the paper's tolerance notation, the method finds a $(\tau, \sqrt{\rho\tau})$ approximate second-order stationary point in

$$\widetilde{O}(\tau^{-5})$$

steps, with potentially substantial dependence on the preconditioner constants, smoothness, objective gap and probability. It uses periodically enlarged steps as part of the saddle-escape argument.

> **Avoid overclaiming:** The paper analyzes particular *controlled* adaptive schemes, not a general proof that every ordinary implementation of Adam or RMSProp converges to local minima from any initialization.

Read Sections 3.1, 4.1–4.2, Definition 4.2, Algorithms 3–4 and Corollary 4.2; inspect the appendix for preconditioner-estimation details.

## 3. What it contributes and what it assumes

**Contributions.** A clean split between the ideal preconditioner and its online estimation; an exact covariance calculation explaining near-isotropic transformed noise; second-order convergence results for a preconditioned stochastic-gradient scheme; a theoretical lens on why adaptivity may improve saddle escape. **Assumptions.** Unbiased stochastic gradients, smooth objective with Hessian regularity, suitable bounded stochastic moments, positive-definite/regularized preconditioner with controlled spectrum and second-moment noise, careful parameter selection and estimation burn-in. **Limitations.** Full-matrix preconditioning is expensive in high dimensions; diagonal alternatives need their own constants and assumptions. The $\widetilde{O}(\tau^{-5})$ bound is not universally superior to SGD. Guarantees are approximate second-order stationarity, not global minima or practical superiority of every adaptive optimizer.

### Connections across the survey

**Daneshmand (2018)** requires natural SGD noise to be correlated with a negative-curvature direction (CNC); Staib et al. use *preconditioning* to make that excitation less direction-dependent. **Jin (2017)** adds external isotropic perturbations; here favorable noise geometry can arise from the optimizer's own gradient statistics. **LENA (Ref 84)** instead controls noisy recursive-gradient estimation during occasional perturbation phases.

### Quick reference

| Concept | Meaning |
|---|---|
| $G = \mathbb{E}[gg^\top]$ | Gradient second moment; equals covariance only at zero mean. |
| $A = (G + \epsilon I)^{-1/2}$ | Preconditioner that equalizes directions when well estimated. |
| $AGA^\top \approx I$ | Noise approximately isotropic near stationarity, if damping small. |
| Burn-in | Gradient-statistics estimation before normal optimization. |
| $\widetilde{O}(\tau^{-5})$ | Second-order convergence order under controlled preconditioner assumptions. |

### Papers Master note

Staib et al. view adaptive methods as preconditioned SGD. Full-matrix inverse-square-root second-moment scaling can whiten intrinsic gradient noise near stationary points, improving saddle-escape directions; a controlled RMSProp-style method gains conditional approximate second-order convergence guarantees.

Source: original ICML 2019 PDF, Sections 3–4, Eq. (3), Definition 4.2, Theorem 4.4 and Corollary 4.2.